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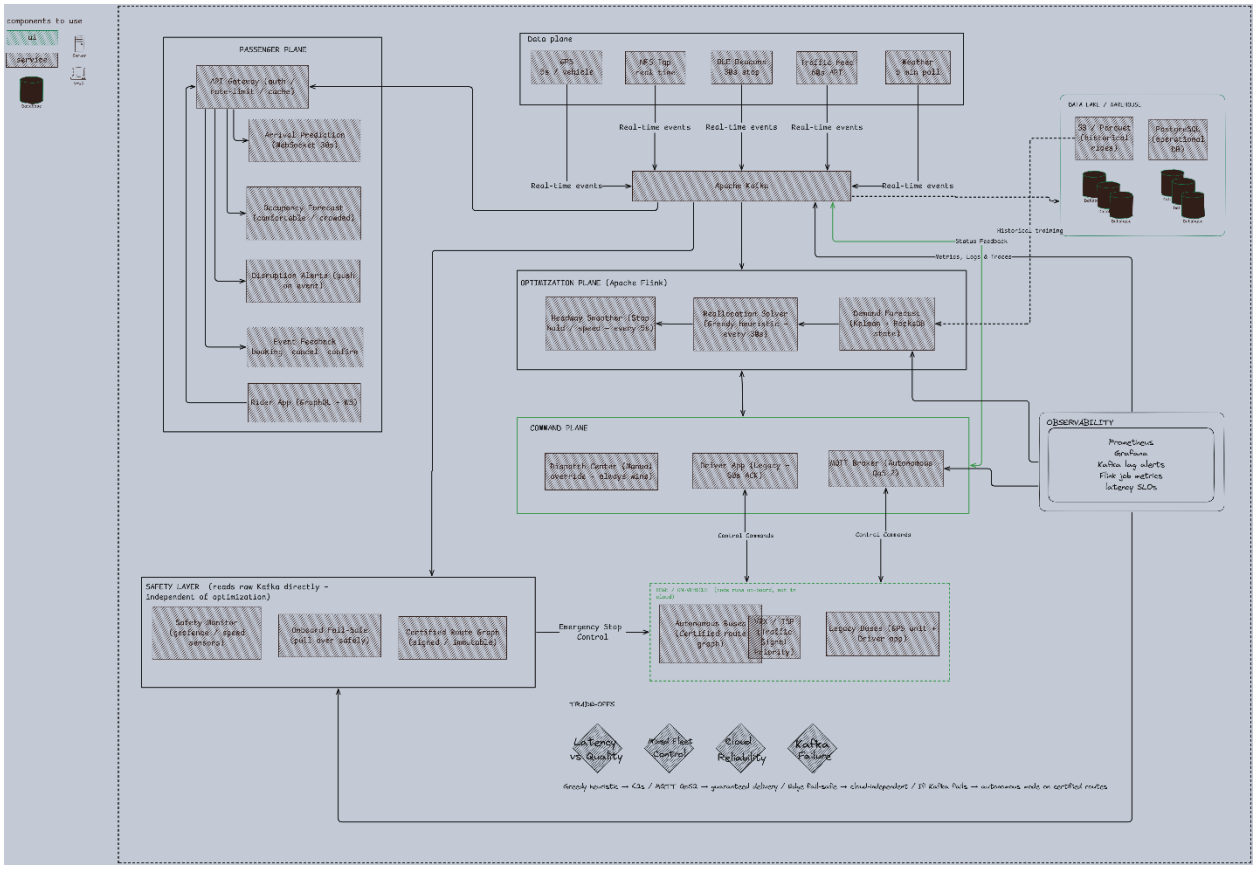
Transportation system design



1. Problem and Assumptions

Buses are overcrowded at peak hours and nearly empty off-peak. Fixed schedules cannot react to real-time demand. The system integrates autonomous vehicles into the existing fleet and adjusts allocation, routes, and schedules in near real-time.

Fleet: 200 legacy buses today, 50 autonomous vehicles in phase 1, up to 200 by phase 3

Latency target: Reallocation decisions reach vehicles within 30 seconds of a triggering event

Availability: 99.95 percent uptime for the dispatch plane; onboard safety systems operate independently of cloud

Regulatory constraint: Autonomous vehicles may only travel pre-certified routes; the fleet system cannot command an uncertified path

2. Architecture Overview

The system has four decoupled planes connected by Apache Kafka topics. A failure in any one plane degrades gracefully without taking down the others.

Data plane: Ingests GPS telemetry every 5 seconds from all vehicles, passenger tap events from NFC validators, stop occupancy from BLE beacons, and city traffic feed updates. All events are published to Kafka partitioned by vehicle ID.

Optimization plane: Apache Flink consumes the data plane topics and runs three nested loops: demand forecasting every 15 minutes, vehicle reallocation every 30 seconds, and headway smoothing every 5 seconds. Solver output is written back to Kafka.

Command plane: Reads solver output and delivers commands to autonomous buses over MQTT with QoS 2, and to legacy bus drivers via a mobile app with a 60-second acknowledgment window.

Passenger plane: Serves arrival predictions, occupancy forecasts, and disruption alerts via WebSocket push to the rider app. Predictions refresh every 30 seconds or immediately on a change exceeding 60 seconds.

3. Optimization Engine

Demand forecasting (every 15 minutes)

A gradient-boosted model trained weekly on 2 years of ridership data produces a baseline forecast per stop per 15-minute slot. A Kalman filter updates the forecast in real-time as observed tap counts diverge from the prediction. A 40 percent demand surge on one route propagates corrections forward to connecting routes within one cycle.

Reallocation solver (every 30 seconds)

A greedy nearest-improvement heuristic assigns vehicles to routes in under 2 seconds, producing solutions within 8 percent of the mathematical optimum. An exact integer program would be optimal but takes 5 to 30 seconds, exceeding the cycle budget. The 8 percent gap is corrected by the next cycle 30 seconds later. Hard constraints: every active route must have at least one vehicle, autonomous vehicles cannot leave their certified route graph, vehicles below 20 percent battery must return to depot.

Headway smoothing (every 5 seconds)

Adjusts holding time at stops and target speed between stops to prevent bus bunching. For autonomous buses commands are executed automatically. For legacy buses the driver app shows a visual hold indicator.

4. Safety Architecture

Safety is enforced at three independent layers. A cloud outage cannot create a safety incident.

Onboard fail-safe: The autonomous vehicle's onboard computer handles all driving decisions independently. If cloud connectivity is lost for more than 10 minutes, the vehicle completes its current stop sequence and pulls over safely to wait for instructions. The fleet system cannot issue driving-level commands, only high-level route assignments.

Certified route graph: Every route an autonomous vehicle may travel must be pre-certified by the transport authority. The certified graph is a signed immutable artifact stored on the vehicle. Any command requiring travel outside the graph is rejected locally before execution, regardless of what the cloud sends.

Safety monitoring service: Runs independently of the optimization engine on the same telemetry stream. Raises alerts for geofence violations above 50 meters, prolonged stops at non-stop locations, speed limit breaches above 5 percent, and sensor fault codes. A dispatcher has 90 seconds to respond; if no action is taken, automatic escalation issues a stop or return-to-depot command directly to the vehicle, bypassing the optimization engine entirely.

Dispatchers retain full override authority. Manual assignments persist until explicitly released and are never reversed automatically by the optimization engine.

5. Key Trade-offs

Optimization quality vs. speed: Greedy heuristic accepted over exact solver. 8 percent suboptimality is tolerable because the 30-second cycle self-corrects. Exact solver would require 5 to 30 seconds, making the system too slow to react to sudden demand spikes.

Automation vs. human control: Dispatchers have contextual knowledge the system lacks. Manual overrides are hard constraints the optimizer cannot reverse, ensuring human judgment always takes precedence without requiring manual approval for routine decisions.

Real-time signals vs. historical forecast: Real-time tap counts are noisy; historical models are slow to adapt. Kalman filter blends both: a single anomalous reading shifts the forecast slightly, a sustained 15-minute deviation shifts it substantially. This prevents both overreaction to noise and underreaction to real events.

Autonomous expansion vs. regulatory caution: Fast deployment would accelerate efficiency gains. Uncertified routes create safety risk and regulatory exposure. The certified route graph constraint is enforced on the vehicle hardware, not only in the cloud, making it impossible to bypass even under operational pressure.

Freshness vs. consistency: Fleet state in Flink is at most 5 seconds stale. Stale state leads to slightly suboptimal assignments but never to unsafe decisions, because safety monitoring reads from the raw telemetry stream independently of the state store.

6. Phased Rollout

Phase 1: Deploy telemetry ingestion and driver app across the full legacy fleet. No autonomous vehicles. Target 15 percent reduction in average wait time within 6 months.

Phase 2: Add 50 autonomous vehicles on 10 certified routes. Enable the reallocation solver. Validate safety incident rate and allocation efficiency.

Phase 3: Introduce demand-responsive micro-routes in low-density zones using smaller autonomous shuttles during off-peak hours.

Phase 4: Scale to 200 autonomous vehicles. Retire legacy buses at end of life rather than replacing with new diesel vehicles.